ASSIGNMENT 4 – PART I AND PART II

SEMANTIC SYSTEMS ENGINEERING:

ANALYSE, REPRESENT, TEST, DISAMBIGUATE AND RETRIEVE

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ABSTRACT

Natural Language Processing systems must go beyond identifying words and grammatical structures. They must represent the meaning conveyed by a sentence, identify the participants in an event, interpret logical relationships, resolve ambiguous words and phrases, and retrieve information according to the user's actual intent. This assignment presents a design-oriented study of semantic systems through two assessment parts. Part I covers event representation, first-order predicate calculus, syntax-driven semantic analysis, word-sense disambiguation, semantic information retrieval and an integrated customer-support scenario. Part II extends these ideas through semantic-role frames, graph representations, quantifier-scope testing, parse-to-meaning procedures, a compact word-sense disambiguation design, scored retrieval traces and an integrated crane-and-bank scenario. Each task follows a traceable workflow: interpret the input, identify semantic units, construct a representation, analyse alternative interpretations, apply a decision procedure, test the result and evaluate limitations. The document uses structured tables, predicate formulae, finite examples and step-by-step reasoning. It does not depend on source code because the assessment emphasises design and analysis.

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PART I – SEMANTIC ANALYSIS AND REPRESENTATION

Part I focuses on the foundations of semantic processing. The six tasks move from a simple event sentence to formal logic, syntactic ambiguity, word-sense disambiguation, semantic retrieval and an integrated support-system example. The answers distinguish what is directly stated from what a system may infer. Where more than one interpretation is possible, alternatives are retained until sufficient evidence is available.

TASK 1 – MEANING REPRESENTATION AND LANGUAGE STRUCTURE

Scenario: A conversational assistant receives the sentence “Maya gave Ravi a book yesterday.” The system must represent who did what, to whom and when.

1.1 Identify Meaning Components

The sentence describes a transfer event. The verb “gave” expresses an action in which an agent transfers an item or theme to a recipient. The noun phrases identify the participants, while “yesterday” locates the event in time. The sentence has a subject–verb–indirect-object–direct-object–time structure.

Component	Expression	Interpretation
Entity 1	Maya	Person who performs the giving event.
Entity 2	Ravi	Person who receives the transferred item.
Entity 3	a book	Object or theme transferred.
Event	gave	A giving or transfer event.
Agent	Maya	Initiator or performer of the event.
Recipient	Ravi	Person who receives the theme.
Theme	a book	The entity transferred.
Time	yesterday	Temporal location of the event.

Linguistic form and conveyed meaning

The linguistic form is the observable arrangement of words: a proper noun, a past-tense verb, two noun phrases and a temporal adverb. It is the surface structure that a parser receives. The

conveyed meaning is the event-level information that the system constructs from those words and their grammatical relations.

The word order helps identify the grammatical subject and objects in this sentence. However, a semantic system should not store only the positions of the words. It should store the roles associated with the event. For example, Maya remains the Agent even if the sentence is rewritten in the passive voice.

The sentence does not explicitly state the book title, its value, the location of the transfer, the reason for giving it or whether Ravi accepted it. A reliable representation must not invent these details. It should represent only the supported facts and leave unknown attributes unspecified.

1.2 Compare Meaning Representations

(a) Conceptual / Event Structure

A simple event representation can be written as:

GIVE(e1) | Agent(e1, Maya) | Recipient(e1, Ravi) | Theme(e1, Book) | Time(e1, Yesterday)

Here, e1 is an event identifier. GIVE identifies the event type. Agent, Recipient, Theme and Time are semantic-role relations connecting the event to its participants and temporal value.

Slot / relation	Value	Why it is included
Event ID	e1	Separates this event from other events in a conversation.
Event type	GIVE	Identifies the action or relation.
Agent	Maya	Identifies who initiates the event.
Recipient	Ravi	Identifies who receives the object.
Theme	Book	Identifies what is transferred.
Temporal value	Yesterday	Anchors the event in time.

(b) Structured Frame / Slot Representation

A frame stores the event as a record with named slots. It is convenient for information extraction, dialogue state tracking and database storage.

Frame: {event_type: GIVE, event_id: e1, agent: Maya, recipient: Ravi, theme: book, tense: past, time_expression: yesterday}.

Frame field	Stored value	Status
event_type	GIVE	Explicit
agent	Maya	Explicit
recipient	Ravi	Explicit
theme	book	Explicit
tense	past	Grammatically indicated
time_expression	yesterday	Explicit
location	unknown	Not stated
purpose	unknown	Not stated
acceptance	unknown	Not stated

Comparison of the two representations

The event structure highlights semantic relations between an event and its participants. It is useful when the same entity may take different roles in different events. The frame representation groups the information into named fields and is easy to store, validate and exchange between system components.

The event representation makes the relationship between event and role explicit. The frame representation makes the values and missing fields easy to inspect. Both preserve the core meaning, but a frame may hide the fact that the slots are relations attached to a particular event unless an event identifier is retained.

Neither representation automatically explains every aspect of natural language. For example, the phrase “a book” is indefinite, but the basic frame does not encode whether the book is new to the conversation. A richer representation could include definiteness, number, reference and discourse status.

1.3 Meaning Structure and Ambiguity

Word order provides grammatical clues, while semantic roles describe the contribution of each participant to the event. In the original sentence, “Maya” is the subject and Agent, “Ravi” is the indirect object and Recipient, and “a book” is the direct object and Theme. The time expression modifies the event.

Meaning-preserving alternative: “Yesterday, a book was given to Ravi by Maya.” The surface word order changes and the sentence becomes passive, but the event roles remain the same.

Equivalent representation: GIVE(e1) ∧ Agent(e1, Maya) ∧ Recipient(e1, Ravi) ∧ Theme(e1, Book) ∧ Time(e1, Yesterday).

A meaning-changing alternative would be “Ravi gave Maya a book yesterday.” This swaps the Agent and Recipient. It is not equivalent because Ravi now performs the action and Maya receives the book.

1.4 Design Defence

The event and frame representations are suitable for a computer system because they convert a sentence into explicit, labelled data. Named roles can be queried, checked and compared without relying on the original word order. The event identifier links every participant to the same action, while the frame makes storage and validation straightforward. Unknown details remain marked as unknown instead of being guessed. The representation can therefore support dialogue systems, information extraction and downstream reasoning.

Task 1 – Summary and Validation

The system should check that the event has one supported Agent, one Theme and the stated Recipient, and that the temporal phrase is attached to the event. It should also preserve the difference between explicit information and inferred or missing information. A validation failure should trigger review rather than silently filling a slot.

TASK 2 – FIRST ORDER PREDICATE CALCULUS

Scenario: A university information system processes statements about students, courses and instructors. First-order predicate calculus (FOPC) is used to express general rules and specific facts precisely.

2.1 Define the Logical Vocabulary

Symbol	Type	Meaning / argument order
Student(x)	Predicate	x is a student.
Course(c)	Predicate	c is a course.
Instructor(i)	Predicate	i is an instructor.
Enrolled(x,c)	Predicate	Student x is enrolled in course c.
Teaches(i,c)	Predicate	Instructor i teaches course c.
Passed(x,c)	Predicate	Student x passed course c.
Rao	Constant	The individual Dr. Rao.
NLP	Constant	The course NLP.
x, c, i	Variables	Range over students, courses and instructors as appropriate.

A predicate is a statement whose truth depends on the values assigned to its arguments. Constants refer to particular entities, while variables range over entities in the domain. The argument order must remain consistent throughout the knowledge base. For example, Enrolled(x,c) always means learner or student first and course second.

2.2 Translate Natural Language into Logic

(i) Every student is enrolled in at least one course

$\forall x\ [\text{Student}(x) \rightarrow \exists c\ (\text{Course}(c) \wedge \text{Enrolled}(x,c))]$

Explanation: For every individual x, if x is a student, there exists at least one individual c that is a course and in which x is enrolled. The existential quantifier is inside the scope of the universal quantifier. Different students may be enrolled in different courses.

(ii) Some student passed every course taught by Dr. Rao

$\exists x\ [Student(x) \wedge \forall c\ ((Course(c) \wedge Teaches(Rao,c)) \rightarrow Passed(x,c))]$

Explanation: There is at least one student x such that, for every course c, if c is a course taught by Dr. Rao, then x passed c. The formula does not require that Dr. Rao teaches at least one course. If Dr. Rao teaches no courses, the universal condition is vacuously true for any student in the domain; an additional existence condition can be added if the intended reading requires a course.

(iii) No instructor teaches every course

$\neg\exists i\ [Instructor(i) \wedge \forall c\ (Course(c) \rightarrow Teaches(i,c))]$

Equivalent form: $\forall i\ [Instructor(i) \rightarrow \exists c\ (Course(c) \wedge \neg Teaches(i,c))]$.

The first form states that there does not exist an instructor who teaches all courses. The equivalent form says that each instructor has at least one course that the instructor does not teach.

2.3 Interpret and Check the Formulae

Consider formula (i): $\forall x\ [Student(x) \rightarrow \exists c\ (Course(c) \wedge Enrolled(x,c))]$. The universal quantifier $\forall x$ has the widest scope. For each possible individual x, the implication checks whether x is a student. If so, the inner existential quantifier searches for at least one course c satisfying Course(c) and Enrolled(x,c). The course may be different for each student.

Quantifier order changes meaning. Compare $\forall x\ \exists c$ Enrolled(x,c) with $\exists c\ \forall x$ Enrolled(x,c). The first says each person may have some course, possibly a different one. The second says there is one particular course in which every person is enrolled.

Finite model	Facts	Formula result
Model A	Students: S1, S2. Courses: C1, C2. Enrolled(S1,C1), Enrolled(S2,C2).	∀x∃c Enrolled(x,c) is true; ∃c∀x Enrolled(x,c) is false.
Model B	Students: S1, S2. Course: C1. Enrolled(S1,C1), Enrolled(S2,C1).	Both formulae are true.

Model A is a counterexample to equivalence. Each student has a course, but there is no single course shared by every student. Therefore, changing the order of the quantifiers changes the claim.

2.4 Logical Consistency

The statements “Every student passed NLP” and “There is a student who did not pass NLP” cannot both be true under the same domain and interpretation. The first statement is ∀x [Student(x) → Passed(x,NLP)]. The second is ∃x [Student(x) ∧ ¬Passed(x,NLP)]. The existential statement identifies a student who violates the universal statement. They are direct logical contradictories, provided the constant NLP refers to the same course in both statements.

Task 2 – Additional Verification Notes

A logic engine can test consistency by attempting to construct a model satisfying all formulae. If the knowledge base contains both the universal passed statement and a fact identifying a student who did not pass NLP, no classical first-order model can satisfy both. The system should report the conflict and identify the relevant facts rather than selecting one statement arbitrarily.

TASK 3 – LINGUISTIC CONCEPTS AND SYNTAX-DRIVEN SEMANTIC ANALYSIS

Input sentence: “The experienced researcher saw the student with the telescope.” The phrase “with the telescope” is structurally ambiguous because it may attach to the verb phrase or to the noun phrase “the student.”

3.1 Identify Linguistically Relevant Concepts

Concept	Sentence expression	Semantic function
Entity	researcher	Person who performs the seeing event.
Entity	student	Person or entity perceived.
Action	saw	Perception event.
Modifier	experienced	Describes the researcher.
Instrument	telescope	Possible instrument used for seeing.
Prepositional phrase	with the telescope	Can modify the seeing event or the student.
Agent	researcher	Experiencer or perceiver.
Theme / stimulus	student	Entity perceived.
Attribute	experienced	Property of the researcher.

These concepts matter because a semantic representation must identify the participants and the relationships between them. The adjective “experienced” modifies the researcher, not the student. The preposition “with” introduces a relation, but the relation depends on its syntactic attachment.

3.2 Syntax-to-Semantics Mapping

Interpretation A: Instrument attachment to the verb phrase

Parse: [The experienced researcher] [saw [the student] [with the telescope]]. In this reading, the prepositional phrase attaches to the verb phrase. The telescope is the instrument used by the researcher to see the student.

Representation: SEE(e1) ∧ Agent(e1, Researcher) ∧ Theme(e1, Student) ∧ Instrument(e1, Telescope) ∧ Experienced(Researcher).

Interpretation B: Modifier attachment to the noun phrase

Parse: [The experienced researcher] [saw [the student with the telescope]]. In this reading, the prepositional phrase attaches to the noun phrase "the student." The student is associated with or carrying the telescope. The sentence does not state that the researcher used the telescope.

Representation: SEE(e1) ∧ Agent(e1, Researcher) ∧ Theme(e1, Student) ∧ With(Student, Telescope) ∧ Experienced(Researcher).

Stage	Verb-phrase attachment	Noun-phrase attachment
Constituent	with the telescope attaches to saw	with the telescope attaches to student
Relation	Instrument(e1, Telescope)	With(Student, Telescope)
Meaning	Researcher uses telescope	Student is associated with telescope
Core event	Researcher sees student	Researcher sees student

Syntax-driven composition procedure

First, tokenise the sentence and identify the noun phrases, verb and prepositional phrase. Next, construct candidate parse trees according to grammatical attachment rules. For each parse, build phrase meanings from the leaves upward. The noun phrase "the experienced researcher"

contributes an entity with the property Experienced. The verb “saw” introduces a perception event with an Agent and a Theme. The phrase “the student” supplies the Theme.

At the final attachment step, the parser combines “with the telescope” with one of two possible constituents. If it combines with the verb phrase, it adds an Instrument role to the event. If it combines with the noun phrase, it adds an association relation to the student. The divergence occurs at the attachment of the prepositional phrase, not at the meaning of the words themselves.

3.3 Semantic Composition

Step-by-step composition

1. Identify lexical meanings: experienced is an adjective; researcher and student are nouns; saw is a past-tense perception verb; with is a preposition; telescope is a noun.
2. Build the subject noun phrase: researcher(x) ∧ Experienced(x).
3. Build the object noun phrase: student(y).
4. Create a perception event: SEE(e1) ∧ Agent(e1,x) ∧ Theme(e1,y).
5. Attach the prepositional phrase. Candidate one adds Instrument(e1,z); candidate two adds With(y,z), where z is the telescope.
6. Keep both representations until discourse context, world knowledge or clarification provides evidence for one reading.

3.4 Failure Analysis

Potential error	Why it occurs	Validation check
Wrong instrument assignment	The parser attaches the phrase to the noun phrase but the semantic module treats it as an event instrument.	Check whether the parse tree attaches the PP to VP; require an Instrument role to have an event as its first argument.
Wrong participant role from word order	The system assumes the nearest noun is always the Agent or Instrument.	Use syntactic dependency relations and verb-specific role constraints; verify that the subject of “saw” is the

		perceiver.

A further risk is that the system may select the most frequent attachment even when the local context supports another interpretation. A robust system should retain an ambiguity record with candidate structures and confidence values. A validation check should ensure that every semantic role is licensed by the selected predicate and that modifiers are attached to the constituent indicated by the parse.

TASK 4 – LEXEMES, SENSES AND WORD SENSE DISAMBIGUATION

The word form “bank” occurs in three contexts: (A) “Nila deposited cash at the bank.” (B) “They sat on the bank of the river.” (C) “The pilot banked the aircraft to the left.” These contexts show that the same spelling can be associated with different meanings and grammatical forms.

4.1 Lexemes and Senses

A lexeme is an abstract vocabulary unit that groups related word forms. A word form is the actual written or spoken form that appears in a sentence. A sense is a particular meaning associated with a lexeme in a context. The noun “bank” in financial and river contexts has different senses. The verb “banked” is a related but grammatically distinct verb form describing an aircraft turning or tilting.

Context	Word form	Part of speech	Relevant sense	Evidence
Nila deposited cash at the bank.	bank	Noun	Financial institution	cash, deposited
They sat on the bank of the river.	bank	Noun	Land beside a river	river, sat on
The pilot banked the aircraft to the left.	banked	Verb, past tense	Tilted or turned an aircraft	pilot, aircraft, to the left

The examples should not be collapsed into one meaning merely because they share the spelling. Grammatical information is also useful: “bank” is a noun in the first two examples, while “banked” is a verb in the third.

4.2 Design a Context-Based WSD Method

The proposed method is a rule-based contextual word-sense disambiguation procedure. It receives a sentence containing a target form from the bank lexeme and returns a sense label with the evidence used.

Design element	Specification
Input	Sentence, target word position and part-of-speech estimate.
Candidate senses	FINANCIAL_INSTITUTION; RIVER_EDGE; AIRCRAFT_TURN.
Context window	Target word plus three tokens on each side; full sentence is also checked.
Features	Nearby verbs, nouns, prepositions, grammatical category and dependency relations.
Decision rule	Score each sense using matching contextual cues; select the highest score if it exceeds a confidence threshold.
Output	Target form, POS, selected sense, evidence words and confidence.

Rule inventory

Sense	Positive cues	Example rule
FINANCIAL_INSTITUTION	cash, deposit, money, account, loan, withdraw	If cash or deposit occurs near noun bank, favour financial sense.
RIVER_EDGE	river, stream, water, shore, sat, grass	If river or water occurs in the local context, favour river-edge sense.
AIRCRAFT_TURN	pilot, aircraft, plane, wing, left, right	If banked is a verb with aircraft or pilot, favour aircraft-turn sense.

The system first identifies the target word and its grammatical category. It then extracts the context window and searches for cue terms. A cue can add a positive score to one or more candidate senses. A strong grammatical cue, such as a past-tense verb form with “aircraft” as an

object, can receive more weight than a weak topical cue. If the top two scores are close, the system should return an unresolved result rather than making an unsupported decision.

4.3 Test the Method

Sentence	Evidence	Selected sense
Nila deposited cash at the bank.	deposited, cash	FINANCIAL_INSTITUTION
They sat on the bank of the river.	river, sat on	RIVER_EDGE
The pilot banked the aircraft to the left.	pilot, aircraft, left; verb form banked	AIRCRAFT_TURN
The bank was covered with reeds beside the stream.	reeds, stream, beside	RIVER_EDGE

Possible failure: “The pilot walked to the bank after landing near the river.” The context contains both pilot and river. If the system uses only a short keyword list, it may choose the aircraft sense because of “pilot,” even though “bank” refers to the river edge. A refinement is to use dependency relations and local syntactic context. In this sentence, “to the bank” is a location phrase, while the aircraft-turn sense normally requires a verb form such as “banked.”

4.4 Evaluation

A small labelled test set can contain 30 sentences, with 10 examples for each of the three senses. Each sentence should be annotated by a human reviewer using the surrounding context. Ambiguous examples should be labelled as ambiguous or excluded from the primary accuracy calculation and reported separately.

Actual sense	Predicted financial	Predicted river	Predicted aircraft	Total
Financial	9	1	0	10
River	1	8	1	10

Aircraft	0	1	9	10
Total	10	10	10	30

Correct predictions = 9 + 8 + 9 = 26. Accuracy = correct predictions / total examples = 26 / 30 = 0.8667, or approximately 86.67%. The confusion matrix also shows the error types: one financial example was labelled river, one river example was labelled financial, one river example was labelled aircraft, and one aircraft example was labelled river.

High accuracy on a balanced test set does not guarantee reliable performance in real text. Real data may contain many more financial uses than aircraft uses, new vocabulary, domain-specific phrases, spelling variation and sentences with insufficient context. The system should therefore be evaluated on data from the intended application domain and should report per-sense precision, recall and F1 in addition to overall accuracy.

TASK 5 – INFORMATION RETRIEVAL AND SEMANTIC MATCHING

Scenario: A library search system receives the query “methods for finding the meaning of ambiguous words” and must retrieve relevant documents.

5.1 Query and Document Representation

The central concepts are method, identify or find, meaning, ambiguous word and context. The likely intent is to find approaches that determine which meaning of a word is intended in a particular sentence. The query is not simply asking for a dictionary definition of ambiguity.

Query concept	Related concepts
Methods	algorithms, procedures, techniques, models
Finding	identify, classify, infer, select
Meaning	sense, interpretation, semantic value
Ambiguous words	polysemy, homonymy, multiple senses
Context	surrounding words, sentence, discourse

A query representation can store the original terms, normalised terms, concept labels and an intent vector. A document representation can store title, abstract, body terms, extracted concepts, domain and semantic embedding. The system should preserve field information so that a relevant title or abstract can receive more weight than an incidental body-word occurrence.

5.2 Lexical versus Semantic Retrieval

Aspect	Keyword matching	Semantic retrieval
Matching basis	Exact or normalised word overlap	Conceptual similarity and contextual meaning
Strength	Fast, transparent and easy to explain	Can connect different wording that expresses the same idea

Weakness	May miss synonyms and paraphrases	May retrieve related but non-answering documents
Example	Finds documents containing “ambiguous words”	May find a document titled “Word Sense Disambiguation in Context”

A relevant document that keyword matching could miss is “A supervised approach to selecting the intended lexical sense from surrounding sentence features.” It may not contain the exact phrase “ambiguous words,” but it directly addresses the user's intent. Semantic retrieval can connect “selecting the intended lexical sense” with “finding the meaning of ambiguous words.”

5.3 Ranking Design

The ranking method combines at least three signals: lexical relevance, concept similarity and field weighting. One possible score is:

$$\text{Score}(q,d) = 0.30L(q,d) + 0.50S(q,d) + 0.20F(q,d)$$

L is a normalised lexical-overlap score between 0 and 1. S is semantic similarity between query and document representations, also normalised between 0 and 1. F is field relevance, giving higher values when key concepts appear in the title or abstract. The weights sum to 1.

Signal	Purpose	Influence
Lexical relevance L	Rewards direct query-term matches.	Helps rank documents that use the user's wording.
Semantic similarity S	Measures whether the document expresses the same concept.	Allows useful paraphrases and synonyms to rank highly.
Field relevance F	Considers where the matching concepts occur.	Promotes documents whose titles or abstracts directly address the query.

5.4 Evaluation and Error Analysis

Assume the system retrieves five documents. Three of them are relevant to the query, while two are related but do not explain methods for identifying word senses. Assume the complete test collection contains four relevant documents.

Retrieved document	Relevant?	Reason
D1: Introduction to Word Sense Disambiguation	Yes	Directly explains the task.
D2: Contextual Embeddings for Lexical Sense Selection	Yes	Explains a semantic method.
D3: Dictionary-Based Sense Identification	Yes	Provides a relevant method.
D4: General History of Dictionaries	No	Related topic but not a method for the query.
D5: Word Frequency in News Articles	No	Shares language-processing vocabulary but does not answer the query.

Precision = relevant retrieved documents / all retrieved documents = 3/5 = 0.60 or 60%. Recall = relevant retrieved documents / all relevant documents in the collection = 3/4 = 0.75 or 75%.

A semantic-retrieval error occurs when a document is conceptually related but does not answer the question. For example, a general dictionary-history article may be semantically close to the word “meaning” but is not a WSD method. The system can reduce this error by adding an answerability or task-intent feature and by giving higher weight to documents that describe an actual method, model or procedure.

TASK 6 – INTEGRATED SEMANTIC ANALYSIS CHALLENGE

Scenario: A customer-support system receives “I saw her duck near the bank, and it was closed.” The system must interpret the message and retrieve the correct support article.

Stage 1 – Interpret the Problem

The sentence contains at least three ambiguities. First, “duck” may be a noun referring to a bird or a verb meaning to lower the head or body. Second, “bank” may refer to a financial institution or the edge of a river. Third, “it” may refer to the bank, a location, or another previously mentioned object. The word “closed” strongly suggests an establishment or a physical object that can be shut, but it does not fully resolve the reference without discourse context.

Stage 2 – Build Candidate Interpretations

Candidate A: Financial institution and bird

Interpretation: The speaker saw a female person's duck near a financial bank building, and the bank was closed. The pronoun “it” refers to the bank.

Representation: SEE(e1) ∧ Agent(e1, Speaker) ∧ Theme(e1, DuckBird) ∧ Near(DuckBird, FinancialBank) ∧ Closed(FinancialBank).

Candidate B: River bank and ducking action

Interpretation: The speaker saw a woman lower her body near the edge of a river, and the location or a nearby facility was closed. This interpretation requires a discourse context that makes the verb “duck” and the pronoun “it” plausible.

Representation: SEE(e1) ∧ Agent(e1, Speaker) ∧ Theme(e1, DuckingEvent) ∧ Agent(DuckingEvent, Woman) ∧ Near(DuckingEvent, RiverBank) ∧ Closed(Referent(it)).

Ambiguous item	Candidate A	Candidate B
duck	Noun: bird	Verb: lower body
bank	Financial institution	River edge

it	Financial institution	Unresolved location or facility
closed	Business not open	Facility or access point unavailable

Stage 3 – Syntax-Driven Analysis and WSD

The parser first identifies the phrase “her duck” as a noun phrase if “duck” is used as a noun. If “duck” is a verb, it expects a different constituent structure, such as “saw her duck” where “her” can be an object pronoun and “duck” a bare infinitive in a perception construction. Both are grammatically possible. The phrase “near the bank” can modify the seeing event or the duck/ducking event depending on the parse.

The phrase “and it was closed” is a separate clause. The pronoun-resolution module should identify possible antecedents and score them using number, grammatical compatibility, discourse salience and the predicate “closed.” A bank as a business is a strong candidate because businesses have opening hours. If evidence is insufficient, the assistant should ask: “Do you mean the financial bank, or the river bank?”

Stage 4 – Information Retrieval Trace

If the context confirms a financial institution, the system builds a query such as: {entity: bank branch, sense: financial institution, attribute: closed, intent: opening hours / branch status}. It searches support articles about branch hours, holiday closures and service availability.

Candidate article	Sense match	Intent match	Decision
A1: How to check bank branch opening hours	High	High	Rank 1
A2: Why river banks close during flood warnings	Low	Low	Reject

A3: How to report a missing pet bird	Duck sense only	Low	Reject
A4: Online banking services when a branch is closed	High	High	Rank 2

The retrieval engine avoids selecting an article about river-bank closures merely because the word “bank” matches. It uses the selected sense, the predicate “closed” and the likely support intent. If the user is asking about a branch being closed, articles about branch opening hours or alternative online services are more useful.

Stage 5 – Challenge and Refine

A misleading context could mention “water” and “branch” near the sentence while the user actually means a bank branch beside a river. A keyword-only system may select the river sense. The refinement is to combine syntactic roles, discourse context and a clarification threshold. The system should retain multiple candidates when the top scores are close and ask a short clarification question before retrieving a potentially incorrect article.

PART II – SEMANTIC SYSTEMS ENGINEERING

Part II extends the first part with more explicit engineering evidence. The design must preserve meaning, expose intermediate decisions, test quantifier scope, challenge syntactically plausible interpretations and evaluate retrieval quality. The answers below use small illustrative datasets and clearly stated assumptions so that the procedures can be reproduced manually.

TASK 1 – REPRESENTING MEANING AND LANGUAGE STRUCTURE

Scenario: A campus assistant receives “Leena sent the report to Arun after the meeting.” The system must capture the event, participants and time relations.

1.1 Semantic Structure Analysis

Semantic unit	Value	Explanation
Event	SEND(e1)	A transfer or communication event.
Agent	Leena	Person who sends the report.
Theme	the report	Information object being sent.
Recipient	Arun	Intended recipient of the report.
Temporal relation	After(e1, Meeting)	Sending occurs after the meeting.
Tense	Past	The event is presented as completed or prior to speech time.
Implicit context	Channel unknown	Email, printed copy or another method is not specified.

The surface form is a declarative sentence in past tense. Its word order places the Agent before the verb, the Theme after the verb and the Recipient in a prepositional phrase. The represented meaning records the event and the semantic roles independently of their surface positions. The phrase “after the meeting” expresses a temporal relation between the sending event and a meeting event.

1.2 Build Meaning Representations

(a) Semantic-role / event frame

SEND(e1) ∧ Agent(e1, Leena) ∧ Theme(e1, Report) ∧ Recipient(e1, Arun) ∧ After(e1, Meeting).

Role	Filler	Evidence
Event type	SEND	Verb “sent”
Agent	Leena	Subject noun phrase
Theme	Report	Direct object
Recipient	Arun	“to Arun”
Temporal relation	After meeting	“after the meeting”

(b) Graph / slot-based representation

A graph representation contains nodes for Leena, the sending event, the report, Arun and the meeting. Directed labelled edges connect the event to its Agent, Theme and Recipient, and connect the event temporally to the meeting.

Source node	Relation	Target node
e1	type	SEND
e1	Agent	Leena
e1	Theme	Report
e1	Recipient	Arun
e1	TemporalRelation	After(Meeting)

A slot-based form is {event: SEND, agent: Leena, theme: report, recipient: Arun, temporal_relation: after, reference_event: meeting}. The graph makes relationships explicit and can connect the event to other events in a larger discourse. The slot frame is compact and convenient for storage. Both leave the communication channel, exact time and delivery confirmation unspecified.

1.3 Meaning-Preserving Variation

Passive alternative: “The report was sent to Arun by Leena after the meeting.” The grammatical subject changes from Leena to the report, but the Agent remains Leena, the Theme remains the report and the Recipient remains Arun. The temporal relation remains After(SendEvent, Meeting).

A meaning-changing alternative is “Arun sent the report to Leena after the meeting.” This reverses Agent and Recipient. The role assignments therefore differ even though the same entities and event type are present.

1.4 Representation Check

Validation rule: every SEND event must have exactly one identified Theme and an Agent; if a Recipient is explicitly present in the input, the Recipient slot must not be empty. The validation module should compare extracted role fillers with dependency relations. If “Arun” is mistakenly assigned as Agent, the mismatch between the grammatical structure and the event-role pattern should raise a review flag.

Task 1 – Additional Engineering Considerations

A real campus assistant may receive fragments such as “Sent the report to Arun after the meeting.” In that case, the Agent may be omitted. The system should not invent Leena as the Agent unless prior dialogue establishes that reference. It can store Agent = unknown and request clarification if the missing role is necessary for the task. This demonstrates the difference between robust representation and unsupported completion.

TASK 2 – FIRST ORDER PREDICATE CALCULUS: FORMALISE AND TEST

Scenario: A course platform stores facts about learners, courses, prerequisites and completion.

2.1 Predicate Vocabulary

Symbol	Meaning	Argument order
Learner(x)	x is a learner.	One argument: learner.
Course(c)	c is a course.	One argument: course.
Enrolled(x,c)	x is enrolled in c.	Learner, course.
Completed(x,c)	x completed c.	Learner, course.
Prerequisite(c1,c2)	c1 is a prerequisite for c2.	Prerequisite course, dependent course.
NLP	Constant for the NLP course.	Specific course.
DepartmentCourse(c)	c is offered by the department.	One argument: course.

Variables x and y range over learners; c, c1 and c2 range over courses. Constants name specific entities. The predicate Prerequisite(c1,c2) is defined as "course c1 must be completed before course c2." This direction must remain fixed so that the formulae are interpretable.

2.2 Formalisation

(i) Every learner enrolled in NLP has completed at least one prerequisite

$\forall x\ [(\text{Learner}(x) \wedge \text{Enrolled}(x,\text{NLP})) \rightarrow \exists c\ (\text{Course}(c) \wedge \text{Prerequisite}(c,\text{NLP}) \wedge \text{Completed}(x,c))]$

The universal quantifier considers every learner. The antecedent restricts the rule to learners enrolled in NLP. For each such learner, there must be at least one course c that is a course, is a prerequisite for NLP and has been completed by that learner.

(ii) Some learner completed every course offered by the department

$\exists x\ [Learner(x) \wedge \forall c\ ((Course(c) \wedge DepartmentCourse(c)) \rightarrow Completed(x,c))]$

There exists at least one learner who completed all courses in the department's course set. The same learner must satisfy the completion relation for every course that meets the condition.

(iii) No learner completed a course that they were not enrolled in

$\forall x \forall c\ [(Learner(x) \wedge Course(c) \wedge Completed(x,c)) \rightarrow Enrolled(x,c)]$

Equivalent form: $\neg \exists x \exists c\ [Learner(x) \wedge Course(c) \wedge Completed(x,c) \wedge \neg Enrolled(x,c)]$. The formula states that there is no learner-course pair for which completion is true and enrolment is false.

2.3 Quantifier-Scope Challenge

Consider $\forall x \exists c$ Completed(x,c) and $\exists c \forall x$ Completed(x,c). The first formula states that every learner completed at least one course, possibly a different course for each learner. The second states that there is one course completed by every learner.

Finite model	Learners and courses	Evaluation
M1	Learners L1,L2; courses C1,C2; Completed(L1,C1), Completed(L2,C2).	$\forall x \exists c$ is true; $\exists c \forall x$ is false.
M2	Learners L1,L2; course C1; Completed(L1,C1), Completed(L2,C1).	Both formulae are true.

M1 proves that the formulae are not logically equivalent. A system that swaps quantifier order could incorrectly convert an individual completion requirement into a shared-course requirement.

2.4 Consistency Test

Illustrative knowledge base:

- Fact 1: Learner(L1).
- Fact 2: Learner(L2).
- Fact 3: Course(NLP).
- Fact 4: Enrolled(L1,NLP).
- Fact 5: Completed(L1,IntroNLP).
- Fact 6: Course(IntroNLP).
- Fact 7: Prerequisite(IntroNLP,NLP).
- Rule: If a learner is enrolled in a course and has completed a prerequisite of that course, the prerequisite requirement is satisfied.

Query: Has L1 completed at least one prerequisite of NLP? The knowledge base contains Course(IntroNLP), Prerequisite(IntroNLP,NLP) and Completed(L1,IntroNLP). Therefore, the existential query $\exists c\ [\text{Course}(c) \wedge \text{Prerequisite}(c,\text{NLP}) \wedge \text{Completed}(\text{L1},c)]$ is true with witness c = IntroNLP.

The query does not follow for L2 because the knowledge base contains no completion fact for L2. Under an open-world assumption, absence of a fact does not prove its negation; the correct answer is “not established,” rather than “false.” If the system uses a closed-world database policy, it may treat missing completion records as false, but that is an additional assumption that must be stated.

TASK 3 – LINGUISTIC CONCEPTS AND SYNTAX-DRIVEN SEMANTIC ANALYSIS

Input sentence: “The analyst noticed the student with the camera near the library.” Two prepositional phrases create attachment ambiguity: “with the camera” and “near the library.”

3.1 Linguistically Relevant Concepts

Concept	Expression	Possible role
Entity	analyst	Agent / perceiver
Entity	student	Theme / noticed entity
Action	noticed	Perception or awareness event
Modifier	the	Determiner for noun phrase
Instrument / association	with the camera	May modify noticing event or student
Location	near the library	May modify event or student
Semantic relation	Agent(e, analyst)	Links event to perceiver
Semantic relation	Theme(e, student)	Links event to noticed entity

The phrase “with the camera” may mean that the analyst used a camera to notice the student, or that the student had a camera. The phrase “near the library” may describe the location of the noticing event or the location of the student. A system must consider both attachment points rather than assuming that each prepositional phrase modifies the nearest noun.

3.2 Parse-to-Meaning Procedure

Parse A: Both phrases modify the event

Structure: [The analyst] [noticed [the student] [with the camera] [near the library]]. The event is located near the library and the camera is used as an instrument.

Representation: NOTICE(e1) ∧ Agent(e1, Analyst) ∧ Theme(e1, Student) ∧ Instrument(e1, Camera) ∧ Location(e1, Near(Library)).

Parse B: Phrases modify the student

Structure: [The analyst] [noticed [the student [with the camera] [near the library]]]. The student is associated with a camera and is near the library. The sentence does not say that the analyst used a camera.

Representation: NOTICE(e1) ∧ Agent(e1, Analyst) ∧ Theme(e1, Student) ∧ Has(Student, Camera) ∧ LocatedNear(Student, Library).

Parse C: Mixed attachment

A third plausible reading is that "with the camera" modifies the event as an instrument, while "near the library" modifies the student. This yields Instrument(e1, Camera) and LocatedNear(Student, Library). The existence of this mixed reading shows why a parser should retain more than one candidate when the syntax permits it.

Attachment decision	Meaning contribution	Candidate relation
with + VP	Camera is used in noticing	Instrument(e1, Camera)
with + NP	Student has/is with camera	Has(Student, Camera)
near + VP	Noticing occurs near library	Location(e1, Near(Library))
near + NP	Student is near library	LocatedNear(Student, Library)

3.3 Composition and Type/Role Checks

Parse-to-meaning trace

7. Tokenise the sentence and identify the noun phrases, verb and prepositional phrases.
8. Build the subject phrase Analyst(x) and the object phrase Student(y).
9. Create the base event NOTICE(e1), with Agent(e1,x) and Theme(e1,y).

10. Generate candidate attachment sites for “with the camera” and add either Instrument(e1,Camera) or Has(y,Camera).
11. Generate candidate attachment sites for “near the library” and add either Location(e1,Near(Library)) or LocatedNear(y,Library).
12. Score the resulting candidate structures using grammar, context and selectional preferences; preserve alternatives if evidence is weak.

Type check 1: Instrument(e,z) must connect an event e to an entity z that can plausibly function as an instrument. It should not connect a person directly to an event if the relation is intended to be Instrument. Type check 2: Agent(e,x) and Theme(e,y) must connect the event to entities of compatible semantic types. The analyst is a plausible perceiver and the student is a plausible perceived entity.

3.4 Error Challenge

A syntactically possible but implausible parse could treat “near the library” as a modifier of the camera, producing a meaning such as “the analyst noticed the student with the camera that was near the library.” This is possible in a larger context, but may be unlikely if the camera is not previously mentioned. The system should use selectional preferences and discourse salience. It can prefer the event-location reading when the surrounding sentences discuss where the analyst observed the student.

The refinement should not hard-code one attachment as always correct. Instead, it should combine syntactic parse probability, semantic type compatibility and contextual evidence. When two candidates remain close, the system should expose the ambiguity or ask a clarification question.

TASK 4 – LEXEMES, SENSES AND WORD SENSE DISAMBIGUATION

Target lexeme: “light” in (A) “Turn on the light.” (B) “This bag is light.” (C) “The light meal was served early.”

4.1 Lexeme and Sense Inventory

The lexeme LIGHT is an abstract vocabulary unit. The word form “light” appears in all three examples, but its part of speech and sense vary. In A it is a noun referring to a source or device that provides illumination. In B it is an adjective meaning low in weight. In C it is an adjective meaning small or not heavy in quantity or richness.

Example	Word form	POS	Sense label	Meaning
Turn on the light.	light	Noun	LIGHT_SOURCE	Device or source that provides illumination.
This bag is light.	light	Adjective	LOW_WEIGHT	Having little weight.
The light meal was served early.	light	Adjective	SMALL / NOT HEAVY MEAL	Not large, rich or heavy.

4.2 WSD Design

A reproducible rule-based method is suitable for this small sense inventory. Input consists of a sentence and the target token. The method first determines the part of speech, then checks local context and grammatical patterns.

Candidate sense	Contextual evidence	Decision rule
LIGHT_SOURCE	turn on, switch on, lamp, bulb, room, illuminate	If light is a noun after a verb such as turn on, select LIGHT_SOURCE.

LOW_WEIGHT	bag, box, carry, weight, heavy, kilogram	If light is a predicate adjective describing an object, select LOW_WEIGHT.
SMALL / NOT HEAVY MEAL	meal, food, breakfast, lunch, snack, rich	If light modifies meal or food, select SMALL / NOT HEAVY MEAL.

Output format: {target: light, POS: noun/adjective, sense: label, evidence: cue list, confidence: high/medium/low}. If no rule matches or two rules match equally strongly, return UNKNOWN or AMBIGUOUS and request more context.

4.3 Apply and Challenge

Sentence	Evidence	Output
Turn on the light.	Imperative verb “turn on”; light is its object.	LIGHT_SOURCE
This bag is light.	Predicate adjective describing bag; contrast with heavy.	LOW_WEIGHT
The light meal was served early.	Adjective immediately modifies meal.	SMALL / NOT HEAVY MEAL
We packed a light lunch for the journey.	light modifies lunch; food context.	SMALL / NOT HEAVY MEAL

Failure case: “The light was too heavy to carry.” Here “light” could refer to a lighting device, while “heavy” indicates physical weight. A simple local cue method may incorrectly select LOW_WEIGHT because it sees “heavy.” The refinement is to use the noun phrase structure and lexical selection: “carry the light” suggests a physical object or device, whereas “the bag is light” describes an object's weight.

4.4 Evaluation

A labelled test set with eight instances can contain multiple examples per sense. Two annotators should label each example according to the intended sense in context. Disagreements should be reviewed and an annotation guide should define the boundaries between a light source, low weight and a light meal.

ID	Test sentence	Gold label	Prediction
1	Turn on the light.	LIGHT_SOURCE	LIGHT_SOURCE
2	The light in the room is broken.	LIGHT_SOURCE	LIGHT_SOURCE
3	This bag is light.	LOW_WEIGHT	LOW_WEIGHT
4	The box feels light.	LOW_WEIGHT	LOW_WEIGHT
5	She prefers a light meal.	SMALL / NOT HEAVY MEAL	SMALL / NOT HEAVY MEAL
6	We ate a light breakfast.	SMALL / NOT HEAVY MEAL	SMALL / NOT HEAVY MEAL
7	The light was difficult to carry.	LIGHT_SOURCE	LOW_WEIGHT
8	The suitcase is light enough to lift.	LOW_WEIGHT	LOW_WEIGHT

There are seven correct predictions out of eight. Accuracy = 7/8 × 100 = 87.5%. The single error is instance 7. If the dataset contained six light-source examples but only one example for each adjective sense, a system that always predicted LIGHT_SOURCE could achieve a deceptively high accuracy. Therefore, the evaluation should include per-class precision, recall and F1, and the test set should represent the range of contexts in actual use.

TASK 5 – SEMANTIC INFORMATION RETRIEVAL

Query: “How can a system identify the intended meaning of a word in context?” The system must retrieve relevant documents from a small collection.

5.1 Query and Document Semantics

The query intent is to learn methods for determining the correct lexical sense of a word based on surrounding context. Key concepts include word, intended meaning, context, sense identification, ambiguity, classification and disambiguation.

Query representation: {task: word-sense disambiguation, target: lexical item, evidence: context, goal: select intended sense, method: algorithm or model}.

Document ID	Candidate document	Relevance label	Reason
D1	A practical guide to Word Sense Disambiguation	Relevant	Directly addresses intended word meaning.
D2	Contextual embeddings for lexical meaning selection	Relevant	Uses context to identify a sense.
D3	Dictionary lookup and lexical sense inventories	Relevant	Provides candidate senses and definitions.
D4	A study of word frequency in news text	Not relevant	Counts words but does not identify intended senses.
D5	The history of printed dictionaries	Not relevant	Shares the concept of word meaning but does not solve contextual ambiguity.

5.2 Ranking Strategy

Use a weighted score combining lexical overlap and semantic similarity:

Score(q,d) = 0.40 × Lexical(q,d) + 0.60 × Semantic(q,d).

Both component scores are normalised to the interval [0,1]. Lexical overlap rewards direct matches such as “word,” “meaning,” “context” and “disambiguation.” Semantic similarity rewards paraphrases such as “selecting a lexical interpretation” even when they do not repeat the query wording. If two documents have equal scores, prefer the document with more direct coverage of the task in its title and abstract.

5.3 Retrieval Trace

Document	Lexical score	Semantic score	Final score
D1	0.90	0.95	0.40(0.90)+0.60(0.95)=0.93
D2	0.55	0.96	0.40(0.55)+0.60(0.96)=0.796
D3	0.70	0.78	0.40(0.70)+0.60(0.78)=0.748
D4	0.65	0.25	0.40(0.65)+0.60(0.25)=0.41
D5	0.40	0.20	0.40(0.40)+0.60(0.20)=0.28

Final ranking: D1, D2, D3, D4, D5. D2 may outrank a keyword-heavy distractor because its contextual-embedding method directly performs the intended task even if its wording differs from the query. D4 contains several general NLP words but does not identify word senses, so its semantic score remains low.

5.4 Evaluate

At cutoff k = 3, the system retrieves D1, D2 and D3. All three are relevant. Precision@3 = 3/3 = 1.00 or 100%. There are three relevant documents in the five-document collection, and all three are retrieved. Recall@3 = 3/3 = 1.00 or 100%.

False positive: D4 may be returned if lexical overlap is given too much weight. False negative: a relevant document that uses only the phrase “contextual lexical interpretation” may be missed if its semantic representation is weak. Improvement: include a query-expansion stage with synonyms and task concepts, then apply a relevance classifier that checks whether the document actually explains a sense-identification method.

TASK 6 – INTEGRATED SEMANTIC ANALYSIS DESIGN CHALLENGE

Scenario: A digital assistant receives “I watched the crane by the bank, then it moved.” The assistant must interpret the sentence and retrieve a suitable information article.

6.1 Model the Ambiguity

The lexeme “crane” may refer to a bird or a machine used for lifting heavy objects. The lexeme “bank” may refer to a financial institution or the land beside a river. The pronoun “it” may refer to the crane or the bank. The verb “moved” is compatible with both a bird and a machine, and may also describe a mobile object or a shifting land feature. Useful context includes whether the scene is a construction site, wetland, river, financial district or industrial area.

6.2 Candidate Meaning Structures

Candidate A: Construction machine near a financial bank

Interpretation: The speaker watched a lifting crane located near a bank building, and the crane moved. “It” refers to the crane.

Representation: WATCH(e1) ∧ Agent(e1, Speaker) ∧ Theme(e1, CraneMachine) ∧ LocatedNear(CraneMachine, FinancialBank) ∧ MOVE(e2) ∧ Theme(e2, CraneMachine) ∧ Before(e1,e2).

Candidate B: Bird near a river bank

Interpretation: The speaker watched a crane bird near the edge of a river, and the bird moved. “It” refers to the bird.

Representation: WATCH(e1) ∧ Agent(e1, Speaker) ∧ Theme(e1, CraneBird) ∧ LocatedNear(CraneBird, RiverBank) ∧ MOVE(e2) ∧ Theme(e2, CraneBird) ∧ Before(e1,e2).

Lexeme	Candidate A sense	Candidate B sense
crane	Lifting machine	Large bird

bank	Financial institution / building	River edge
it	Crane machine	Crane bird
moved	Machine changed position	Bird changed position

6.3 Syntax and WSD Trace

The parser identifies “the crane” as the object of “watched” and “by the bank” as a prepositional phrase that most naturally gives the location of the crane or watching event. The clause “then it moved” introduces a pronoun. The antecedent resolver considers crane and bank. The pronoun is singular, so both candidates are grammatically possible. The predicate “moved” is compatible with both a bird and a machine, so this clue alone is insufficient.

Contextual cues can resolve the senses. Words such as “construction,” “lifting,” “steel” and “operator” favour a crane machine. Words such as “wetland,” “wings,” “nest” and “marsh” favour a crane bird and river bank. If these cues are absent, the assistant should retain both interpretations or ask whether the user means a bird or a construction machine.

6.4 Semantic Retrieval

Assume the context confirms a construction machine near a financial bank. The query representation becomes {entity: crane machine, relation: located near, location: financial bank, event: moved, intent: information about crane movement / construction equipment}.

Document	Content	Sense match	Rank decision
R1	Safety procedures for moving mobile construction cranes	High	Rank 1: matches machine and movement.
R2	How cranes migrate near river banks	Low	Reject: bird sense.
R3	Bank branch locations and opening	Low	Reject: financial bank keyword only; does not answer crane

	hours		movement.
R4	Operating a lifting crane near buildings	High	Rank 2: matches machine and nearby building context.

The system should not select R2 simply because it contains both “crane” and “bank.” It must check the intended sense of each lexeme and the event “moved.” R1 and R4 have the strongest combination of sense compatibility and task relevance.

6.5 Adversarial Refinement

A misleading context could contain the phrase “construction cranes gather near the river bank,” where “river bank” is the intended location but “construction cranes” are machines. A model that uses one dominant topic word may incorrectly select the financial-bank sense. The refinement is to disambiguate each target lexeme independently using local syntax and nearby evidence, then combine the selected senses in the full sentence representation. The system should also keep confidence values and request clarification if the evidence is insufficient.

OVERALL CONCLUSION

This assignment demonstrates how semantic processing converts natural language into explicit meaning structures that can be tested and used by computer systems. Part I introduced event roles, structured frames, first-order predicate calculus, syntactic attachment, lexeme and sense distinctions, word-sense disambiguation and semantic information retrieval. Part II extended these concepts into engineering procedures with graph representations, quantifier-scope countermodels, parse-to-meaning traces, sense inventories, ranked retrieval examples and an integrated ambiguity challenge.

A key lesson is that surface word order alone is not enough to represent meaning. Systems must identify semantic roles and relations, define logical vocabulary carefully, preserve quantifier scope and distinguish a word's form from its contextual sense. When a phrase or word has multiple interpretations, the system should build candidate representations and use syntax, context, semantic compatibility and discourse information to select among them.

Evaluation is essential. Accuracy, precision and recall provide measurable evidence, but a single score does not prove that a system will work reliably in every domain. Balanced test sets, clear annotation rules, representative examples, adversarial cases and error analysis are needed. Semantic similarity also does not guarantee truth or relevance; retrieval systems must verify that a document answers the user's intent.

The complete engineering workflow is therefore: interpret the input, identify semantic units, choose a representation, analyse meaning, resolve ambiguity, retrieve or infer an answer, test the result, examine failure cases, refine the design and defend the final decision. A reliable semantic system should be able to explain not only what answer it produced, but also how it reached that interpretation and where uncertainty remains.

REFERENCES AND ASSUMPTIONS

The answers are based primarily on the supplied assessment brief and standard concepts from Natural Language Processing, formal semantics, first-order logic, word-sense disambiguation

and information retrieval. The document uses illustrative finite models, example document collections, scores and test-set outcomes to demonstrate the required calculations. These examples are constructed for explanation and are not claimed to be experimental results from a real deployed system.

- Jurafsky, D. and Martin, J. H. Speech and Language Processing. Concepts used: lexical semantics, word-sense disambiguation, semantic roles and information retrieval.
- Allen, J. Natural Language Understanding. Concepts used: semantic representation, parsing and interpretation.
- Russell, S. and Norvig, P. Artificial Intelligence: A Modern Approach. Concepts used: first-order logic, knowledge representation and inference.
- Manning, C. D., Raghavan, P. and Schütze, H. Introduction to Information Retrieval. Concepts used: precision, recall and ranking.
- Assessment brief supplied for Natural Language Processing Assignment 4 – Part I and Part II.

FINAL SUBMISSION CHECKLIST

- All six Part I tasks are answered.
- All six Part II tasks are answered.
- Semantic units and participant roles are identified.
- Event, frame, graph and predicate representations are included.
- Quantifier scope and quantifier-order differences are explained.
- Alternative syntactic attachments are represented.
- Lexemes, word forms and senses are distinguished.
- WSD procedures include inputs, features, decision rules and outputs.
- Retrieval examples include ranking signals and calculations.

- Precision, recall and accuracy are calculated from stated examples.
- Ambiguities, assumptions, failure cases and refinements are discussed.
- Source code is not included because the assessment permits documented manual analysis.